

Substitution Parameters from Age-Verification Event Studies

1 Setup

Goods. We work with seven goods: six adult content categories — PornHub (PH), xHamster (XH), xVideos (XV), XNXX, Chaturbate (CB), and a residual Other-XXX bundle (OXXX) — plus an outside option (AO) covering all other uses of time. Let $\mathcal{A} = \{\text{PH}, \text{XH}, \text{XV}, \text{XNXX}, \text{CB}, \text{OXXX}\}$. AO is constructed residually:

$$m_{\text{AO},t}^m = 10080 - \sum_{j \in \mathcal{A}} m_{j,t}^m,$$

where $10080 = 7 \times 24 \times 60$ is the number of minutes in a week. AO is the “left-out” good whose share is pinned by adding-up; we write equations only for the six adult goods.

Time shares. The outcome of interest is the time share

$$s_i^{mt} = \frac{m_i^{mt}}{10080},$$

for machine m , week t , site i . Because 10080 is total available time (not total browsing time), the adult shares are small — $s_A = \sum_{j \in \mathcal{A}} s_j \ll 1$ — and AO absorbs nearly all of the budget. This is a modeling choice; an alternative denominator is total browsing time, which would yield a different scaling of the Slutsky parameters but identical diversion ratios.

Access costs. Each site carries a shadow access cost p_i incorporating compliance burden, user friction, and voluntary withdrawal. Age-verification laws raise the access cost of affected sites. The resulting change in access cost, $\Delta \ln p_i$, may differ across sites and is not directly observed; what we observe is the change in usage Δs_i .

Notation. $s_0 \equiv s_{\text{AO}} = 1 - s_A$, $s_A \equiv \sum_{j \in \mathcal{A}} s_j$.

2 The General Model: AIDS

2.1 Specification

The Almost Ideal Demand System (Deaton and Muellbauer, 1980) relates changes in time shares to changes in log access costs. Suppressing the total-time term — we treat $T = 10080$ as fixed — the linearized system is

$$\Delta s_i = \sum_{j=1}^6 \gamma_{ij} \Delta \ln p_j, \quad i \in \mathcal{A}. \tag{1}$$

The 6×6 matrix $\Gamma = [\gamma_{ij}]$ is the Slutsky substitution matrix for time shares. Three restrictions apply:

$$\begin{aligned} \text{Symmetry: } & \gamma_{ij} = \gamma_{ji} \\ \text{Adding-up: } & \sum_{i=0}^6 \gamma_{ij} = 0 \quad \forall j \quad (\text{sum includes AO}) \\ \text{Homogeneity: } & \sum_{j=1}^6 \gamma_{ij} = 0 \quad \forall i. \end{aligned}$$

Together these reduce the 6×6 system to $\binom{6}{2} = 15$ free parameters.

2.2 Interpretable objects

We do *not* normalize the diagonal. The change in access cost $\Delta \ln p_j$ for any shocked site j is unobserved, but this is not a problem for the objects we care about.

Diversion ratios. When site j 's access cost rises by $\Delta \ln p_j$, site j 's own share falls by $|\gamma_{jj}| \cdot \Delta \ln p_j$ and site i 's share rises by $\gamma_{ij} \cdot \Delta \ln p_j$. The fraction of j 's lost share that flows to i is

$$DR_{j \rightarrow i} = -\frac{\gamma_{ij}}{\gamma_{jj}}. \quad (2)$$

The unknown $\Delta \ln p_j$ cancels. Slutsky symmetry ($\gamma_{ij} = \gamma_{ji}$) does *not* imply symmetric diversion ratios: $DR_{j \rightarrow i}/DR_{i \rightarrow j} = |\gamma_{ii}|/|\gamma_{jj}|$, which equals one only when own-price effects are equal.

Cross-price elasticities. $\varepsilon_{ij} = \gamma_{ij}/s_i$ measures the percentage-point change in site i 's share per unit increase in $\ln p_j$, normalized by i 's baseline share.

Allen–Uzawa elasticity. $\sigma_{ij} = \gamma_{ij}/(s_i s_j)$ is a symmetric measure of substitutability; under nested logit it equals $(1/\lambda - s_0)/s_A$ for pairs in the same nest.

Adding-up applies to diversion ratios: summing across all sites including AO,

$$\sum_{i \neq j} DR_{j \rightarrow i} + DR_{j \rightarrow \text{AO}} = 1,$$

so diversion to adult substitutes plus diversion to the outside option sum to one. A sum less than one in the data indicates either declining total time (the fixed- T assumption is violated) or measurement error.

3 The Simple Model: Nested Logit

3.1 Utility and choice probabilities

We adopt the McFadden (1978) nested logit with two nests: the adult nest $\mathcal{A}$ and the singleton outside option. Consumers draw utility

$$U_i = V_i + \varepsilon_i, \quad i \in \mathcal{A}, \quad U_0 = V_0 + \varepsilon_0,$$

where the ε_i for $i \in \mathcal{A}$ follow a generalized extreme value (GEV) distribution with dissimilarity parameter $\lambda \in (0, 1]$. Standard nested logit yields conditional and marginal share equations:

$$s_{i|A} = \frac{\exp(V_i/\lambda)}{\sum_{k \in \mathcal{A}} \exp(V_k/\lambda)}, \quad i \in \mathcal{A}, \quad (3)$$

$$s_A = \frac{D_A^\lambda}{1 + D_A^\lambda}, \quad D_A = \sum_{k \in \mathcal{A}} \exp(V_k/\lambda). \quad (4)$$

The unconditional share is $s_i = s_{i|A} \cdot s_A$.

3.2 Interpretation of λ

λ is the *dissimilarity parameter* (also called the nesting coefficient). It governs the degree of within-nest correlation in unobserved utility: lower λ means higher correlation and stronger preference clustering within the adult nest.

- $\lambda = 1$: standard multinomial logit. Independence of irrelevant alternatives (IIA) holds globally. Adult sites substitute for each other no more than for non-adult time.
- $\lambda \rightarrow 0$: within-nest correlation approaches one. Adult sites are nearly perfect substitutes; a shock to one adult site sends almost all diverted demand to other adult sites.
- $\lambda = 0.217$ (our estimate): strong within-nest preference correlation. Adult sites are substantially closer substitutes for each other than for the outside option, but imperfect.

Define the composite substitution scalar:

$$\varphi_\lambda \equiv \frac{1}{\lambda} - s_0 \approx \frac{1 - \lambda}{\lambda} \quad (\text{since } s_0 \approx 1). \quad (5)$$

φ_λ replaces the $\phi \equiv \sigma_2 - \sigma_1$ combination from a nested CES. Like ϕ , it measures excess within-adult substitution beyond the IIA baseline: $\varphi_\lambda = 0$ when $\lambda = 1$ (IIA), and $\varphi_\lambda \rightarrow \infty$ as $\lambda \rightarrow 0$.

3.3 Slutsky matrix

Linearizing the nested logit share equations around the baseline yields the following Slutsky share derivatives:

Pair	Formula	Sign ($\lambda < 1$)
$i \neq j$, both $\in \mathcal{A}$	$\gamma_{ij} = \frac{s_i s_j}{s_A} \varphi_\lambda$	> 0
$i \in \mathcal{A}$, $j = \text{AO}$	$\gamma_{i,0} = s_i s_0$	> 0
$i \in \mathcal{A}$ (own price)	$\gamma_{ii} = -\frac{s_i}{\lambda} + \frac{s_i^2}{s_A} \varphi_\lambda$	< 0

These satisfy symmetry and adding-up. The within-adult cross-price effects γ_{ij} have the same $s_i s_j / s_A$ factorization as in nested CES, with φ_λ playing the role of ϕ .

3.4 Diversion ratios under nested logit

Applying equation (2) with $i \neq j$, both $\in \mathcal{A}$:

$$DR_{j \rightarrow i} = \frac{(s_i s_j / s_A) \varphi_\lambda}{(s_j / \lambda) - (s_j^2 / s_A) \varphi_\lambda} = \frac{s_i \varphi_\lambda}{s_A / \lambda - s_j \varphi_\lambda}. \quad (6)$$

The numerator is proportional to s_i ; the denominator does not depend on i . This yields the same testable proportionality restriction as nested CES:

Within-adult proportionality. For any $i, k \in \mathcal{A} \setminus \{j\}$,

$$\frac{DR_{j \rightarrow i}}{DR_{j \rightarrow k}} = \frac{s_i}{s_k}. \quad (7)$$

Relative diversion from a shutdown of site j to competing adult sites is proportional to their pre-ban time shares.

Equation (7) holds regardless of λ , the shock magnitude, or which adult site shuts down.

Diversion to the outside option:

$$DR_{j \rightarrow AO} = \frac{s_A s_0}{s_A / \lambda - s_j \varphi_\lambda}. \quad (8)$$

The ratio of total within-adult diversion to outside-option diversion identifies φ_λ :

$$\frac{\sum_{i \neq j, i \in \mathcal{A}} DR_{j \rightarrow i}}{DR_{j \rightarrow AO}} = \frac{(s_A - s_j) \varphi_\lambda}{s_A s_0} \approx \frac{1 - \lambda}{\lambda}, \quad (9)$$

where the approximation uses $s_0 \approx 1$ and $s_j \ll s_A$. The right-hand side involves only the single primitive λ , directly identified from the data without knowing shock magnitudes.

4 Identification Given Simultaneous Shocks

4.1 Setup

Age-verification laws generate three simultaneous access-cost shocks, each of unknown magnitude:

1. **PH:** PornHub's access cost rises by $\Delta \ln p_{PH} > 0$ (unobserved).
2. **XH:** xHamster's access cost rises by $\Delta \ln p_{XH} > 0$ (unobserved), roughly simultaneously with PH.
3. **OXXX:** Some sites within the Other-XXX bundle face the same mandate; the bundle's effective access cost rises by $\Delta \ln p_{OXXX} > 0$ (unobserved).

All three shocks are of unknown magnitude. We do not observe the price changes; we observe only the resulting changes in time shares Δs_i . The event study for site i estimates the composite shock:

$$\Delta s_i = \gamma_{i,PH} \cdot \Delta \ln p_{PH} + \gamma_{i,XH} \cdot \Delta \ln p_{XH} + \gamma_{i,OXXX} \cdot \Delta \ln p_{OXXX} + \varepsilon_i. \quad (10)$$

This holds for every site $i \in \mathcal{A} \cup \{AO\}$, including the three directly shocked sites.

4.2 Diversion ratios cancel the unknown shock magnitudes

Suppose geographic variation allows us to isolate the shock to site j : for states where only j faces the mandate, the event study yields $\hat{\beta}_i^j \approx \gamma_{ij} \cdot \Delta \ln p_j$ for each site i . The diversion ratio is then

$$\widehat{DR}_{j \rightarrow i} = \frac{\hat{\beta}_i^j}{|\hat{\beta}_j^j|} = \frac{\gamma_{ij} \cdot \Delta \ln p_j}{|\gamma_{jj}| \cdot \Delta \ln p_j} = \frac{\gamma_{ij}}{|\gamma_{jj}|}, \quad (11)$$

and $\Delta \ln p_j$ cancels.

4.3 Problem 1: PH and XH are simultaneously shocked

If PH and XH are always banned in the same states at the same time, the event study produces a single combined coefficient $\hat{\beta}_i^{\text{combined}} \approx \gamma_{i,\text{PH}} \cdot \Delta \ln p_{\text{PH}} + \gamma_{i,\text{XH}} \cdot \Delta \ln p_{\text{XH}}$. We cannot separately identify the PH and XH columns of the Slutsky matrix; only their shock-weighted average is estimable.

What nested logit buys here. Under nested logit, PH and XH are in the same nest with the same within-adult proportionality structure. The ratio of diversion to XV versus XNXX satisfies $DR_{j \rightarrow \text{XV}}/DR_{j \rightarrow \text{XNXX}} = s_{\text{XV}}/s_{\text{XNXX}}$ for $j \in \{\text{PH}, \text{XH}\}$ — the same for both shocked sites. Thus the proportionality test (7) and the identification of λ from equation (9) remain valid under the combined shock.

What geographic variation buys. States that ban only PH (not XH) and states that ban only XH (not PH) separately identify the PH and XH columns of the Slutsky matrix.

4.4 Problem 2: OXXX is simultaneously shocked

The OXXX shock contaminates the reduced-form coefficients for sites that didn't face the mandate directly. Under nested logit, $\gamma_{i,\text{OXXX}} = (s_i s_{\text{OXXX}}/s_A) \varphi_\lambda$, which scales with s_i for all $i \in \{\text{XV}, \text{XNXX}, \text{CB}\}$. The OXXX term therefore adds the same proportional bias to all three non-shocked adult sites. Because the ratios $\Delta s_{\text{XV}}/\Delta s_{\text{XNXX}}$ are unaffected by this proportional contamination, the proportionality test (7) and the ratio in equation (9) remain valid. The *level* of φ_λ (and hence λ) is, however, biased upward by OXXX contamination.

4.5 Summary: what is identified

Object		Without geo. variation	With geo. variation
Proportionality $DR_{j \rightarrow i} / DR_{j \rightarrow k} = s_i / s_k$	test	Yes (nested logit prediction holds under combined shock)	Yes
Dissimilarity parameter λ		Yes, directly identified (from ratio of within-adult to AO diversion)	Yes
λ in levels (not derived combination)		Yes (single primitive, no normalization needed)	Yes
PH vs. XH columns separately (AIDS)		No (only shock-weighted sum identified)	Yes (PH-only and XH-only states)
9 of 15 AIDS parameters		No	Yes (PH and XH columns via Slutsky symmetry)

5 Estimation with the Machine-Week Panel

5.1 Data structure

The existing analysis uses a stacked TWFE panel of machine $m \times$ cohort $k \times$ week t observations, with dependent variable `win_min`: winsorized minutes per machine per week, by site. There are approximately 14.4 million machine-week observations across 14 state cohorts. For the demand system, we construct

$$s_i^{mtk} = \frac{\text{win_min}_{i,mtk}}{10080}, \quad s_0^{mtk} = 1 - \sum_{j \in \mathcal{A}} s_j^{mtk}.$$

5.2 Option 1: Two-step minimum distance

Step 1. Run the stacked TWFE event study for each site j separately, including separate treatment indicators D_{stk}^j for each shocked site ($j \in \{\text{PH, XH, OXXX}\}$), leveraging geographic variation in which sites are banned and when. This yields reduced-form coefficients $\hat{\beta}_i^j$ for each outcome site i and each shocked site j .

Step 2. Compute diversion ratios and structural parameters. The diversion ratio from j to i follows equation (11). The nested logit identification equation is

$$\frac{\beta_i}{\beta_{\text{AO}}} = \frac{s_i}{s_A} \cdot \frac{1 - \lambda}{\lambda}, \quad (12)$$

solving directly for λ :

$$\lambda = \frac{1}{1 + \frac{\beta_i}{\beta_{\text{AO}}} \cdot \frac{s_A}{s_i}}. \quad (13)$$

The proportionality restriction (7) is testable by comparing $\hat{\beta}_i^j / \hat{\beta}_k^j$ to s_i / s_k .

5.3 Option 2: Seemingly Unrelated Regressions (SUR)

Stack all six adult site share equations in a joint system:

$$s_{mtk}^i = \alpha_{mk}^i + \alpha_{tk}^i + \sum_{j \in \mathcal{S}} \gamma_{ij} \cdot D_{stk}^j + \varepsilon_{mtk}^i, \quad (14)$$

where $\mathcal{S} = \{\text{PH, XH, OXXX}\}$. Estimate jointly imposing:

- *Slutsky symmetry*: γ_{ij} in equation i equals γ_{ji} in equation j .
- *Nested logit*: $\gamma_{ij} = (s_i s_j / s_A) \varphi_\lambda$ for $i \neq j \in \mathcal{A}$, collapsing the 15-parameter system to **one parameter** λ (not two as in nested CES).

The SUR uses the full panel variation and allows formal likelihood-ratio or Wald tests of the nested logit restrictions against the unrestricted AIDS.

Precision from the panel. Machine fixed effects α_{mk}^i absorb time-invariant taste differences. Within-machine pre/post variation is the key signal of substitution.

Heterogeneity across user types. The SUR framework provides a natural common scale for comparing substitution patterns across user groups. In the SUR, all equations use time shares, so the estimated λ and implied diversion ratios are dimensionless and directly comparable across groups.

5.4 Heterogeneous nested logit

Let $g \in \mathcal{G}$ index a person type, and let $g(m)$ denote the type of machine m . The type-specific SUR is:

$$s_{mtk}^{ig} = \alpha_{mk}^{ig} + \alpha_{tk}^{ig} + \sum_{j \in \mathcal{S}} \gamma_{ij}^g \cdot D_{stk}^j + \varepsilon_{mtk}^{ig}. \quad (15)$$

Under the type-specific nested logit, the structural Slutsky parameters are:

$$\gamma_{ij}^g = \frac{s_i^g s_j^g}{s_A^g} \varphi_\lambda^g, \quad \varphi_\lambda^g = \frac{1 - \lambda^g}{\lambda^g}, \quad i \neq j \in \mathcal{A}, \quad (16)$$

where s_i^g is the pre-ban time share for type g and λ^g is the type- g dissimilarity parameter. The identification equation becomes

$$\lambda^g = \frac{1}{1 + \frac{\beta_i^g}{\beta_{AO}^g} \cdot \frac{s_A^g}{s_i^g}}, \quad (17)$$

directly identifying λ^g for each group from existing heterogeneity event studies and group-specific pre-ban shares.

What the type heterogeneity captures. Two sources of between-type variation enter equation (16):

1. *Structural heterogeneity*: $\lambda^g \neq \lambda$. Different types have genuinely different within-nest preference correlation. Heavy users may have lower λ^g (stronger within-adult correlation) while occasional users have λ^g closer to 1.

2. *Composition heterogeneity*: even if $\lambda^g = \lambda$ for all g , the Slutsky matrix Γ^g varies across types because baseline shares s_i^g differ.

Testable restrictions.

- H_0 : $\lambda^g = \lambda \forall g$ — dissimilarity parameter is homogeneous across types. Test via Wald test on the interaction of D_{stk}^j with type dummies in the SUR.
- H_0 : $DR_{j \rightarrow i}^g / DR_{j \rightarrow k}^g = s_i^g / s_k^g$ — within-adult proportionality holds for type g .

Example economic hypotheses.

- Heavy past-XXX users likely have low λ^g : strong preferences over the adult nest mean within-adult sites are close substitutes.
- Occasional users likely have λ^g closer to 1: adult consumption is more easily displaced to the outside option.
- Users with children in the household may have intermediate λ^g generally with similar within-adult correlation when they do access content.